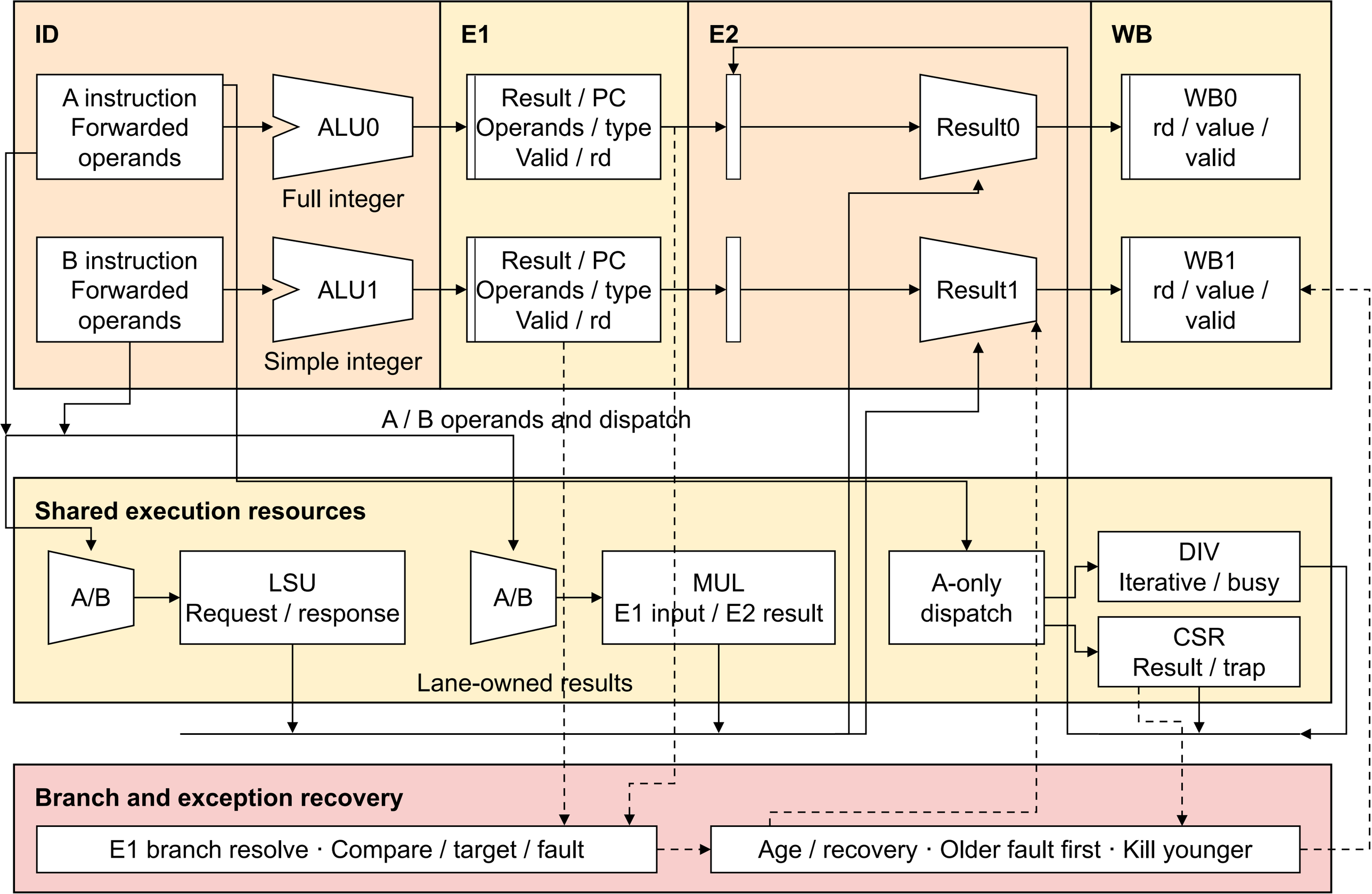
ID
E1
E2
WB
A instruction
Forwarded
operands
ALU0
Full integer
Result / PC
Operands / type
Valid / rd
Result0
WB0
rd / value /
valid
B instruction
Forwarded
operands
ALU1
Simple integer
Result / PC
Operands / type
Valid / rd
Result1
WB1
rd / value /
valid
A / B operands and dispatch
Shared execution resources
A/B
LSU
Request / response
A/B
MUL
E1 input / E2 result
A-only
dispatch
DIV
Iterative / busy
CSR
Result / trap
Lane-owned results
Branch and exception recovery
E1 branch resolve · Compare / target / fault
Age / recovery · Older fault first · Kill younger